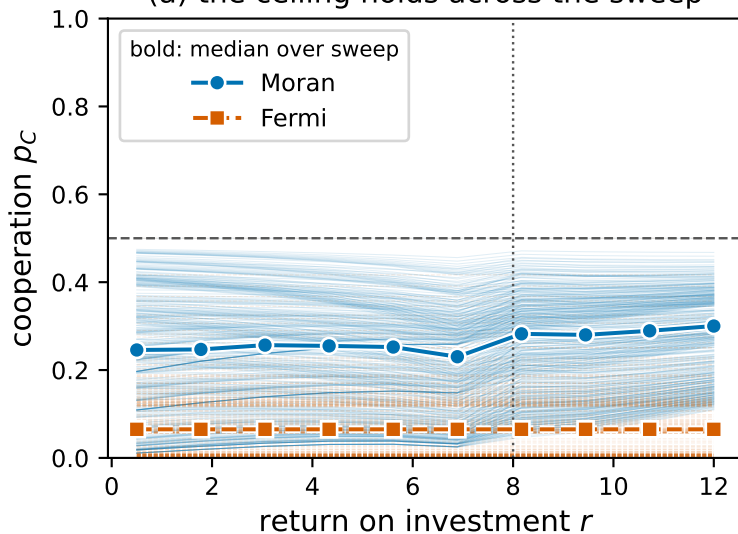
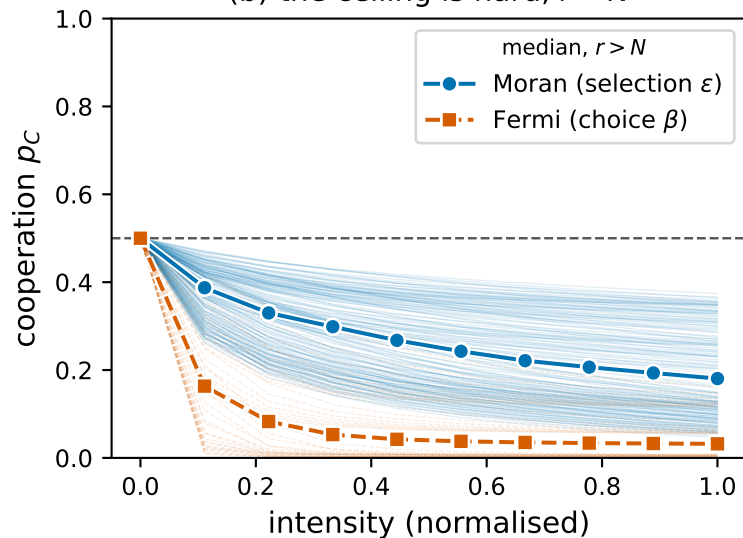


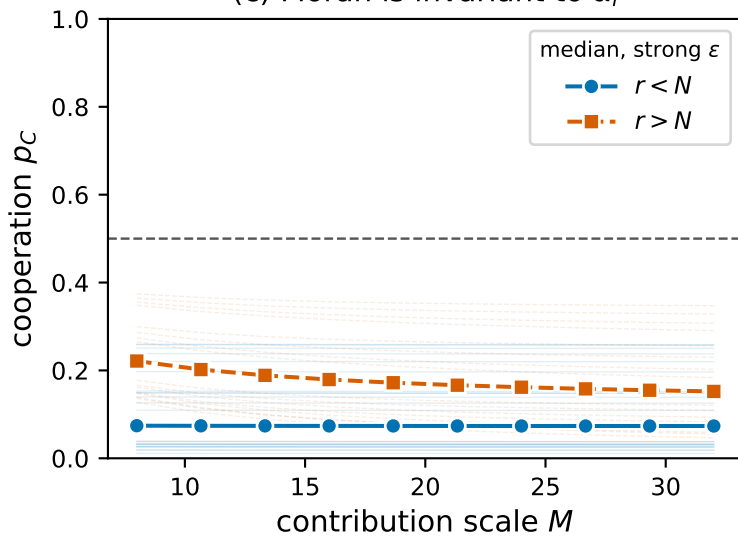
(a) the ceiling holds across the sweep



(b) the ceiling is hard, $r > N$



(c) Moran is invariant to α_i



(d) Fermi is invariant to r

